Climate-Shock Energy Risk

How Gulf hurricanes and US winter storms move oil and gas prices, and whether risk models see them coming · EIA data to 2026-09-22 · generated 2026-09-28

Headline numbers

EVENTS TESTED

18 + 11

Gulf hurricanes + winter storms with a full year of prior data

GAS VOLATILITY AFTER WINTER STORMS

1.58x

of the month before; same-season days 0.81x (q = 0.077)

OIL VOLATILITY AFTER HURRICANES

0.99x

same-season days 0.96x (p = 0.751)

HENRY HUB REGIME SHIFT, WINTER

+0.28

change in Pr(turbulent); placebo −0.07

PLAIN HS VAR IN STORMS (GAS)

3.4%

breach rate vs 0.9% on other days (target 1%)

WORST CLIMATE SCENARIO, BOOK

−$893k

Winter storm Dec 2022; 10-day VaR $850k

- **Winter storms are a gas-volatility event; the average move hides a fat tail.** In the ten trading days after a winter storm, Henry Hub volatility averaged 1.58x its level in the month before, against 0.81x on random days in the same months (placebo p 0.005, FDR q 0.077). Most storms barely register: the average is driven by Winter storm Jan 2024 (15x), Winter storm Feb 2021 (13x), Winter storm Jan 2018 (7x).
- **Hurricanes do not move oil the way headlines suggest.** Across 18 Gulf hurricanes, crude volatility after landfall was 0.99x the month before (same-season days: 0.96x; p 0.751), and no average price move survives a same-season placebo test after correcting for 16 tests (smallest q 0.22).
- **A regime model notices winter storms in real time.** A two-state Markov-switching model fitted only on data to 2014 and then run as a live filter raised its probability of the turbulent regime for Henry Hub by +0.28 after winter storms, against −0.07 on placebo dates (q 0.003). It shows no such shift for hurricanes in any series.
- **The VaR model that fails in storms is the one that does not update volatility.** Plain historical simulation breached on 3.4% of storm-window days for a long gas position, against 0.9% elsewhere (Fisher p <0.001). Filtered HS and GARCH-t, which re-weight to recent volatility every day, stayed near 1.2% (p 0.306).
- **A climate overlay learned only from past storms cuts storm-window breaches cheaply.** For the desk book it takes filtered-HS breaches in storm windows from 5 to 2 and GARCH-t from 6 to 2, for 1.1% and 1.4% more average VaR. It cannot rescue plain HS in the extreme cold snaps, which lost up to 4x VaR in a day.
- **Stress test:** replaying each storm's 13-day price moves on today's book, the worst case is Winter storm Dec 2022 at −$893k (1 of 30 scenarios exceed today's 10-day 99% VaR of $850k). The worst 13 days of the whole sample, ending 2020-03-30, cost −$1.73mm: climate is a real but not the dominant source of tail risk for this book.

> **Note.** Events are picked by physical rules (a storm's track and winds; a disaster's damage), never by what prices did, and every test compares events with random days from the same calendar months, because hurricane season and winter are volatile anyway.

> **Caveat.** This build used fallback copies of the event data (hurricane: https://raw.githubusercontent.com/tidyverse/dplyr/main/data-raw/storms.csv; winter: bundled snapshot of NCEI events-US-1980-2024.csv) because the NOAA servers were unreachable. `bash run.sh` re-downloads the official files.

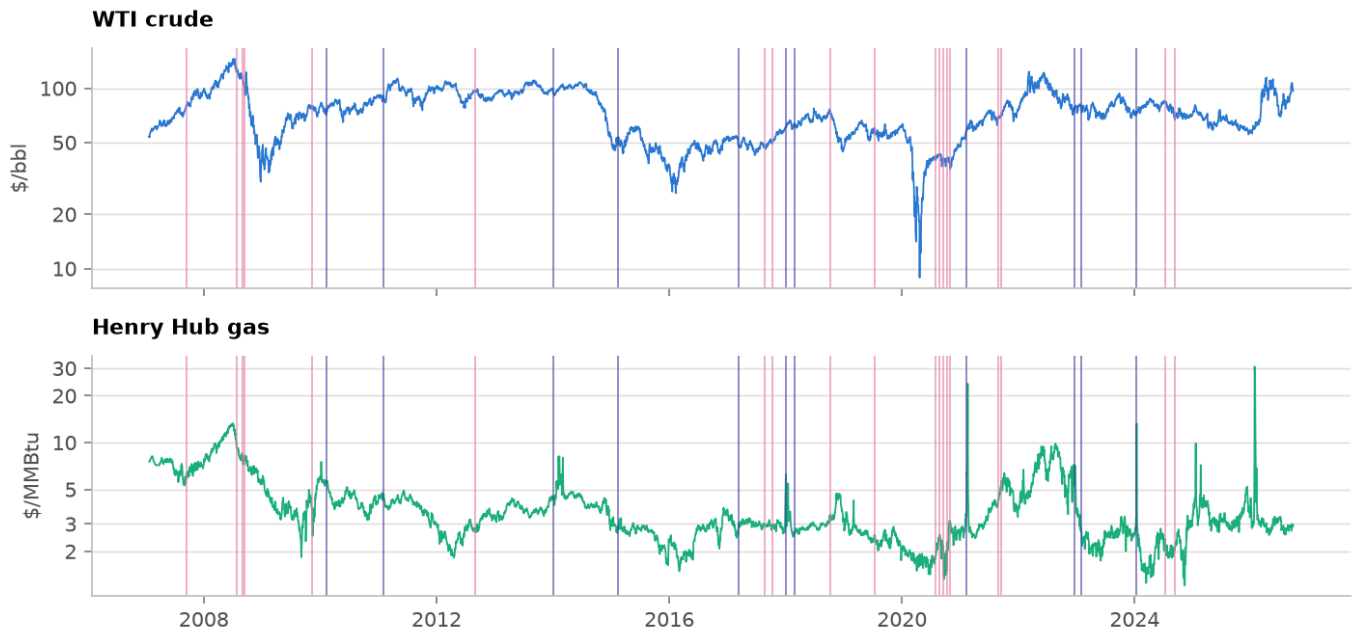
1. The events

Hurricanes come from NOAA's best-track database (HURDAT2): a storm counts if it is at hurricane strength (at least 64 knots) while inside the northern Gulf box 25.5-30.5°N, 98.0-86.0°W, which covers the federal offshore oil and gas fields and the Texas-to-Alabama refining coast. Day 0 is the first date it is there. **Winter storms** come from NOAA NCEI's billion-dollar disaster list: every "Winter Storm" or "Freeze" that began between November and March, with episodes less than a week apart merged. Day 0 is the first trading day on or after the event date.

Where the events come from

Hazard	Source	Data through	Events	Rule
hurricane	https://raw.githubusercontent.com/tidyverse/dplyr/main/data-raw/storms.csv	2024-11-18	19	status HU, >= 64 kt, lat (25.5, 30.5), lon (-98.0, -86.0)
winter	bundled snapshot of NCEI events-US-1980-2024.csv	2024-01-14	12	Winter Storm or Freeze, starting in months (11, 12, 1, 2, 3), merged within 7 days

19 Gulf hurricanes (pink) and 11 winter storms (purple), picked by physical rules



Every event on the WTI and Henry Hub price history.

The event list

Event	Hazard	Date	Day 0	Detail
Humberto 2007	Gulf hurricane	2007-09-13	2007-09-13	max 80 kt in the Gulf box, 3 six-hourly fixes
Dolly 2008	Gulf hurricane	2008-07-23	2008-07-23	max 85 kt in the Gulf box, 6 six-hourly fixes
Gustav 2008	Gulf hurricane	2008-08-31	2008-09-02	max 95 kt in the Gulf box, 6 six-hourly fixes
Ike 2008	Gulf hurricane	2008-09-11	2008-09-11	max 95 kt in the Gulf box, 10 six-hourly fixes

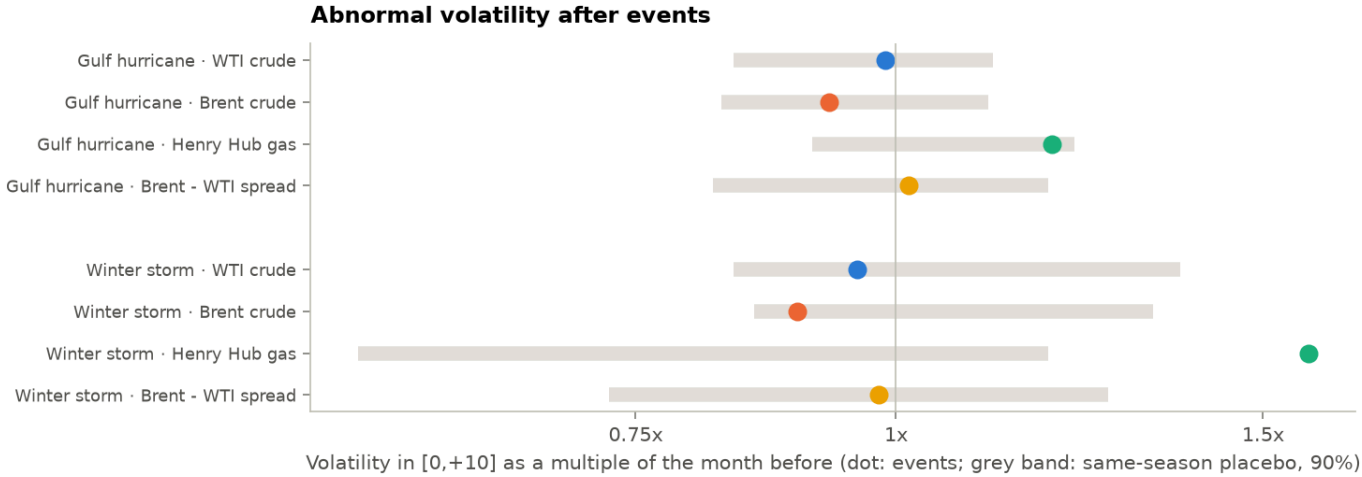
Event	Hazard	Date	Day 0	Detail
Ida 2009	Gulf hurricane	2009-11-09	2009-11-09	max 75 kt in the Gulf box, 2 six-hourly fixes
Winter storm Feb 2010	Winter storm / freeze	2010-02-09	2010-02-09	NCEI billion-dollar event, Feb 09-Feb 11 2010
Winter storm Feb 2011	Winter storm / freeze	2011-02-01	2011-02-01	NCEI billion-dollar event, Feb 01-Feb 03 2011
Isaac 2012	Gulf hurricane	2012-08-28	2012-08-28	max 70 kt in the Gulf box, 7 six-hourly fixes
Winter storm Jan 2014	Winter storm / freeze	2014-01-05	2014-01-06	NCEI billion-dollar event, Jan 05-Jan 08 2014
Winter storm Feb 2015	Winter storm / freeze	2015-02-14	2015-02-16	NCEI billion-dollar event, Feb 14-Feb 20 2015
Freeze Mar 2017	Winter storm / freeze	2017-03-14	2017-03-14	NCEI billion-dollar event, Mar 14-Mar 16 2017
Harvey 2017	Gulf hurricane	2017-08-25	2017-08-25	max 115 kt in the Gulf box, 7 six-hourly fixes
Nate 2017	Gulf hurricane	2017-10-07	2017-10-09	max 80 kt in the Gulf box, 4 six-hourly fixes
Winter storm Jan 2018	Winter storm / freeze	2018-01-03	2018-01-03	NCEI billion-dollar event, Jan 03-Jan 05 2018
Winter storm Mar 2018	Winter storm / freeze	2018-03-01	2018-03-01	NCEI billion-dollar event, Mar 01-Mar 03 2018
Michael 2018	Gulf hurricane	2018-10-09	2018-10-09	max 125 kt in the Gulf box, 4 six-hourly fixes
Barry 2019	Gulf hurricane	2019-07-13	2019-07-15	max 65 kt in the Gulf box, 2 six-hourly fixes
Hanna 2020	Gulf hurricane	2020-07-25	2020-07-27	max 80 kt in the Gulf box, 5 six-hourly fixes
Laura 2020	Gulf hurricane	2020-08-26	2020-08-26	max 130 kt in the Gulf box, 5 six-hourly fixes
Sally 2020	Gulf hurricane	2020-09-14	2020-09-14	max 95 kt in the Gulf box, 10 six-hourly fixes
Delta 2020	Gulf hurricane	2020-10-09	2020-10-09	max 105 kt in the Gulf box, 5 six-hourly fixes
Zeta 2020	Gulf hurricane	2020-10-28	2020-10-28	max 100 kt in the Gulf box, 4 six-hourly fixes
Winter storm Feb 2021	Winter storm / freeze	2021-02-10	2021-02-10	NCEI billion-dollar event, Feb 10-Feb 19 2021
Ida 2021	Gulf hurricane	2021-08-28	2021-08-30	max 130 kt in the Gulf box, 7 six-hourly fixes
Nicholas 2021	Gulf hurricane	2021-09-14	2021-09-14	max 65 kt in the Gulf box, 3 six-hourly fixes
Winter storm Dec 2022	Winter storm / freeze	2022-12-21	2022-12-21	NCEI billion-dollar event, Dec 21-Dec 26 2022
Winter storm Feb 2023	Winter storm / freeze	2023-02-02	2023-02-02	NCEI billion-dollar event, Feb 02-Feb 05 2023
Winter storm Jan 2024	Winter storm / freeze	2024-01-12	2024-01-12	NCEI billion-dollar event, Jan 12-Jan 18 2024
Beryl 2024	Gulf hurricane	2024-07-08	2024-07-08	max 80 kt in the Gulf box, 3 six-hourly fixes
Francine 2024	Gulf hurricane	2024-09-11	2024-09-11	max 90 kt in the Gulf box, 6 six-hourly fixes

2. Price response: an event study

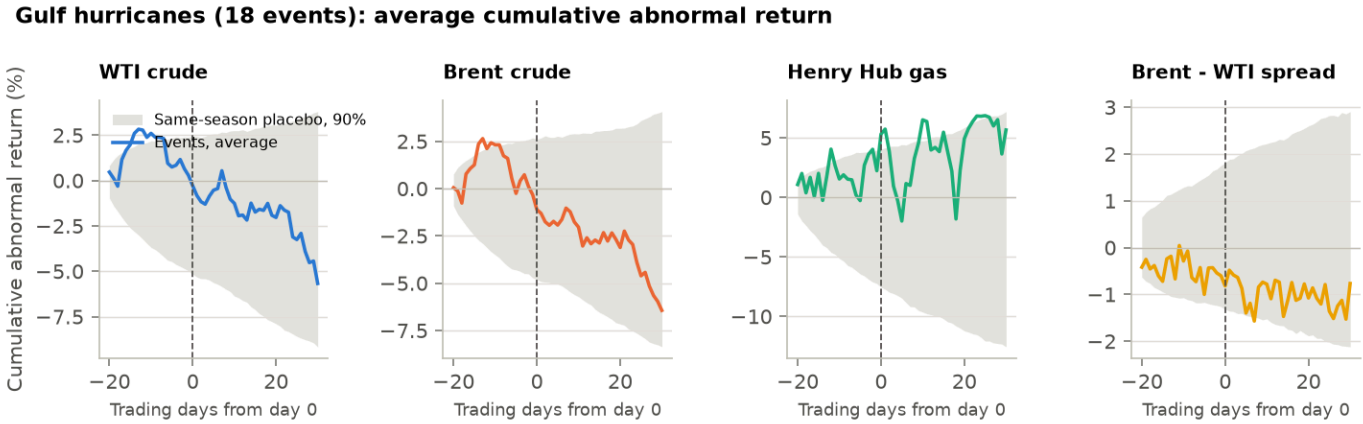
For each event, the normal return is the average over trading days -250 to -21. Abnormal returns are deviations from it, summed into a cumulative abnormal return (CAR) for direction. Abnormal volatility compares the ten days after day 0 with the month before, for size. Each average is tested three ways: a t-test across events, a sign test, and a placebo that repeats the calculation on 5,000 random sets of dates from the same calendar months with no event within 30 trading days. Benjamini-Hochberg q-values control the false-discovery rate across the 16 primary tests.

Primary tests: CAR over [-5,+10] and abnormal volatility over [0,+10]

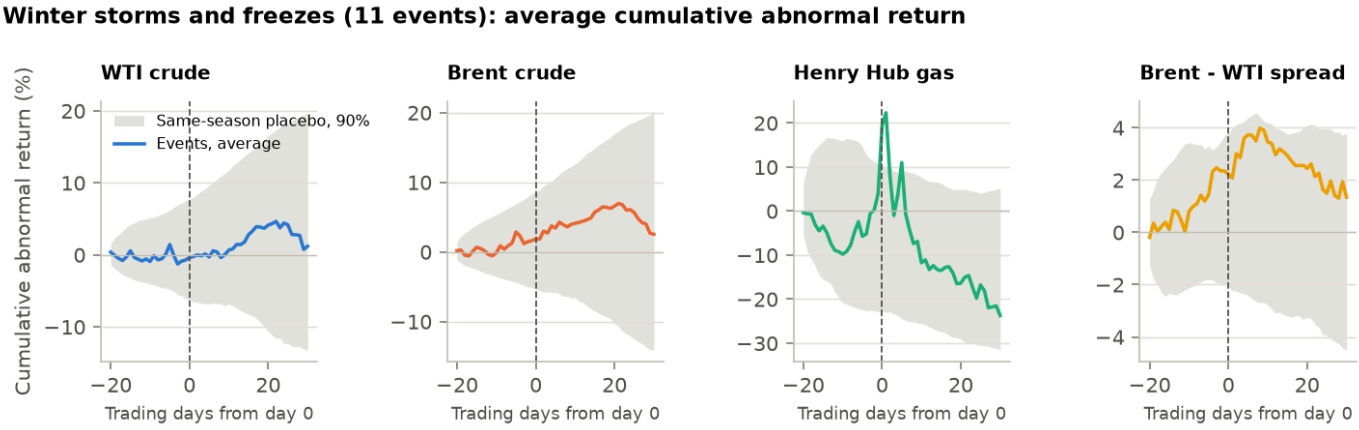
Hazard	Series	Measure	Events	Placebo	t-test p	Placebo p	q (FDR)
Gulf hurricane	WTI crude	Full CAR	-2.21%	-0.71%	0.328	0.430	0.845
Gulf hurricane	WTI crude	Abnormal volatility	0.99x	0.96x	0.917	0.751	0.869
Gulf hurricane	Brent crude	Full CAR	-2.56%	-0.46%	0.308	0.260	0.693
Gulf hurricane	Brent crude	Abnormal volatility	0.93x	0.95x	0.477	0.787	0.869
Gulf hurricane	Henry Hub gas	Full CAR	+6.31%	-1.11%	0.200	0.028	0.221
Gulf hurricane	Henry Hub gas	Abnormal volatility	1.19x	1.05x	0.261	0.174	0.653
Gulf hurricane	Brent - WTI spread	Full CAR	-0.35%	+0.25%	0.663	0.454	0.845
Gulf hurricane	Brent - WTI spread	Abnormal volatility	1.01x	0.98x	0.914	0.753	0.869
Winter storm / freeze	WTI crude	Full CAR	+0.57%	+2.09%	0.808	0.642	0.869
Winter storm / freeze	WTI crude	Abnormal volatility	0.96x	1.05x	0.501	0.581	0.869
Winter storm / freeze	Brent crude	Full CAR	+2.84%	+2.17%	0.283	0.950	0.950
Winter storm / freeze	Brent crude	Abnormal volatility	0.90x	1.06x	0.235	0.204	0.653
Winter storm / freeze	Henry Hub gas	Full CAR	-9.34%	-4.77%	0.307	0.476	0.845
Winter storm / freeze	Henry Hub gas	Abnormal volatility	1.58x	0.81x	0.292	0.005	0.077
Winter storm / freeze	Brent - WTI spread	Full CAR	+2.26%	+0.08%	0.160	0.176	0.653
Winter storm / freeze	Brent - WTI spread	Abnormal volatility	0.98x	0.95x	0.853	0.815	0.869



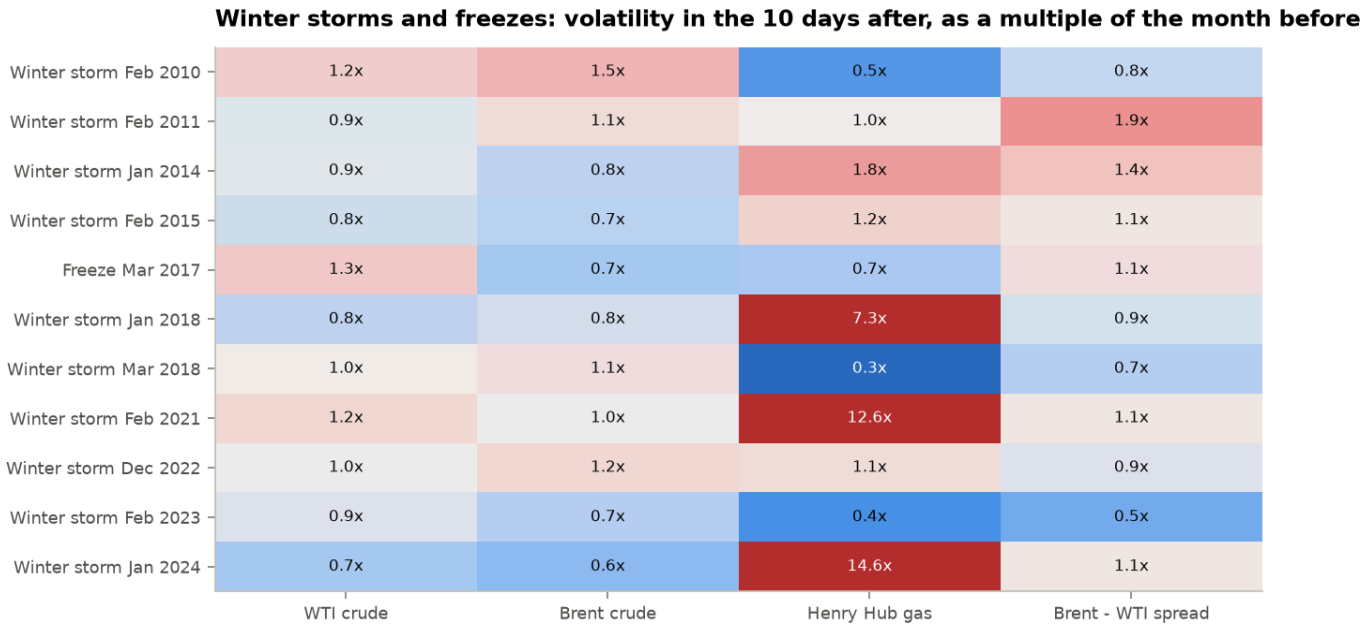
Volatility after events as a multiple of the month before, against the same-season placebo range.



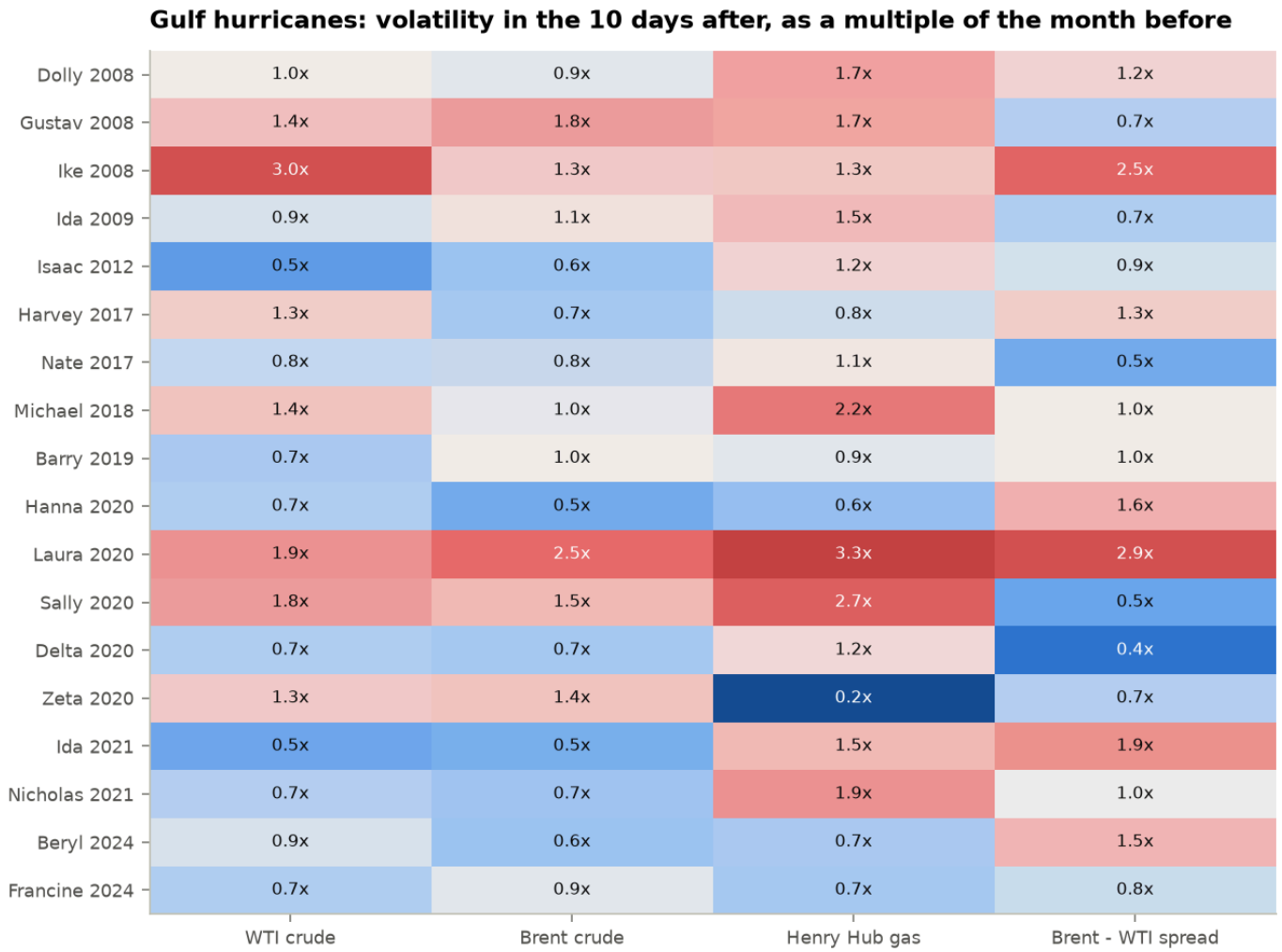
Hurricanes: average cumulative abnormal return; grey = 90% range of same-season placebo sets.



Winter storms: average cumulative abnormal return.



Winter storms, one row each: most barely register; a few multiply gas volatility.



Hurricanes, one row each.

3. Do regime models notice?

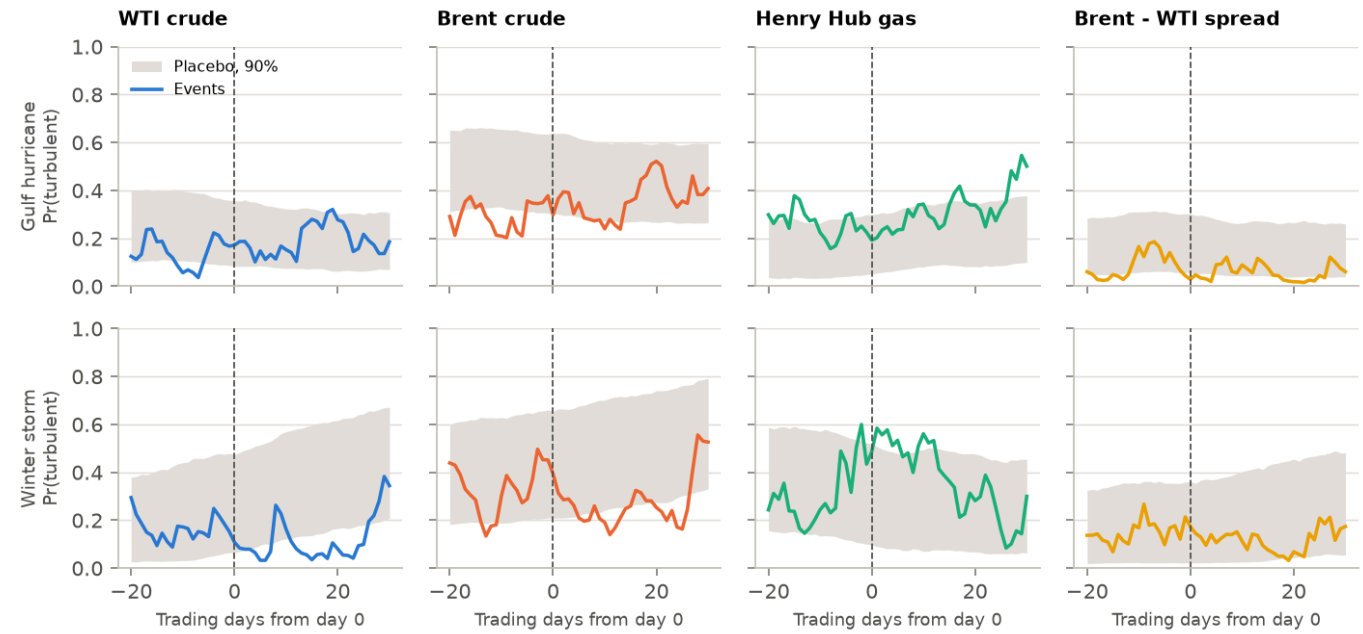
regime_risk's two-state Markov-switching model is fitted on 2007-2014 only, then run forward as a real-time Hamilton filter with fixed parameters. The shift is the average probability of the turbulent regime over days 1 to +10 minus days -15 to -6, for events after the training period.

Fitted regimes (annualised volatility, expected spell length)

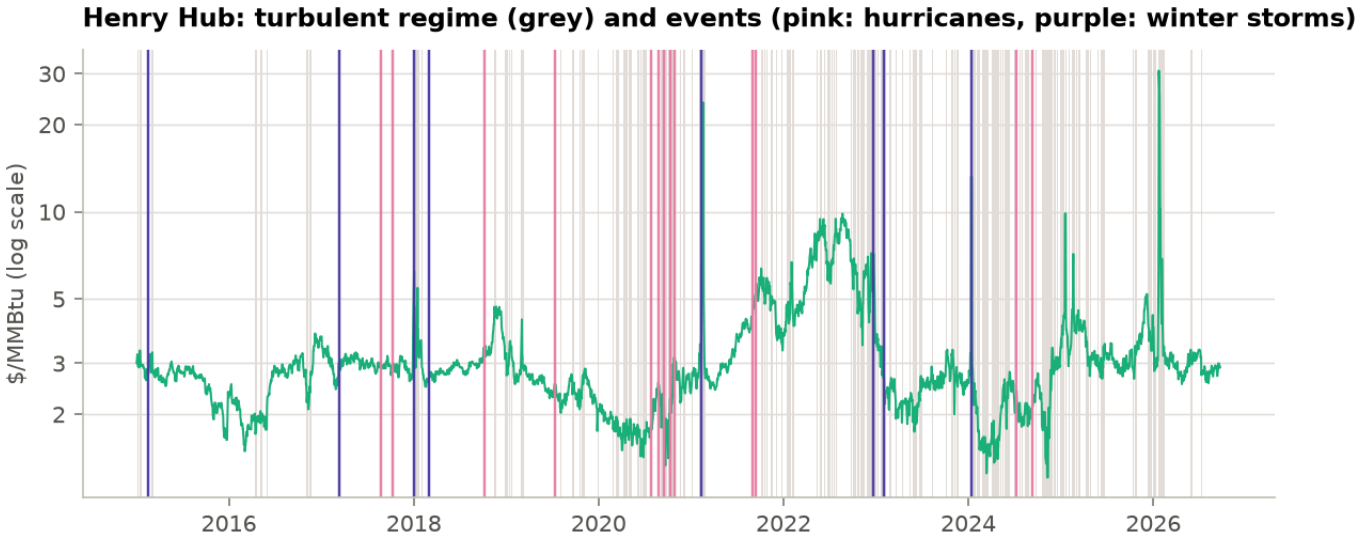
Series	Calm vol	Turbulent vol	Calm spell (days)	Turbulent spell (days)
WTI crude	25%	77%	79	13
Brent crude	21%	53%	48	18
Henry Hub gas	42%	155%	67	8
Brent - WTI spread	20%	67%	38	8

Change in the real-time probability of the turbulent regime

Hazard	Series	Events	Shift	Placebo shift	p	q (FDR)
Gulf hurricane	WTI crude	13	+0.04	−0.04	0.253	0.517
Winter storm / freeze	WTI crude	8	−0.03	+0.08	0.241	0.517
Gulf hurricane	Brent crude	13	+0.06	−0.04	0.259	0.517
Winter storm / freeze	Brent crude	8	−0.02	+0.04	0.551	0.682
Gulf hurricane	Henry Hub gas	13	+0.01	+0.04	0.559	0.682
Winter storm / freeze	Henry Hub gas	8	+0.28	−0.07	<0.001	0.003
Gulf hurricane	Brent - WTI spread	13	−0.04	−0.02	0.682	0.682
Winter storm / freeze	Brent - WTI spread	8	−0.03	+0.02	0.635	0.682



Average real-time probability of the turbulent regime around events, with the placebo range.



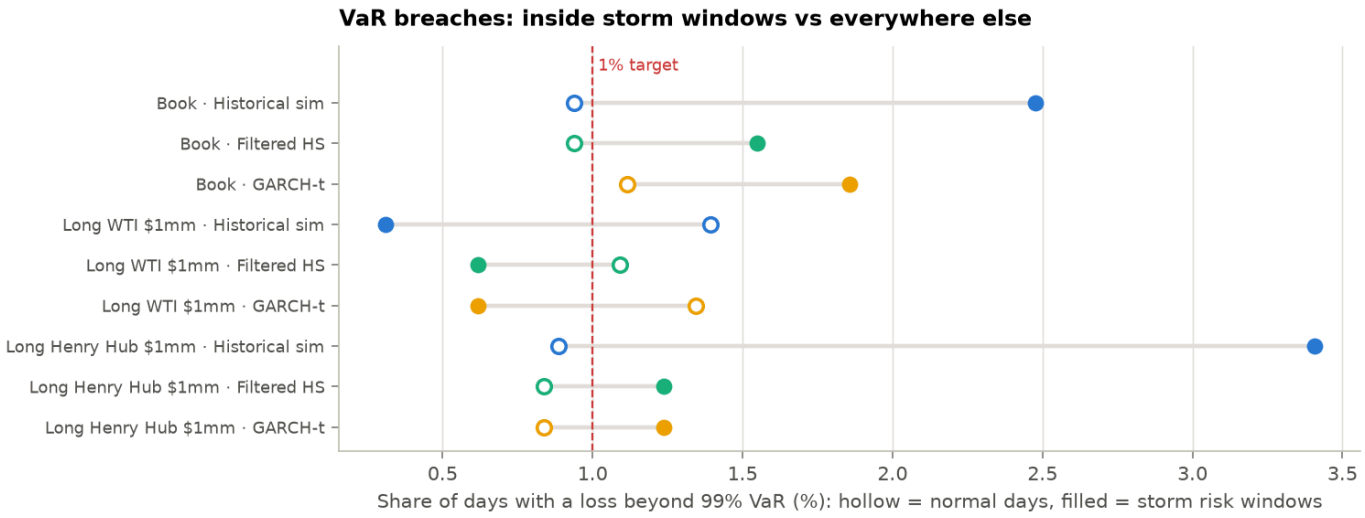
Henry Hub since 2015: turbulent regime (grey) and events.

4. VaR in storms, and a climate overlay

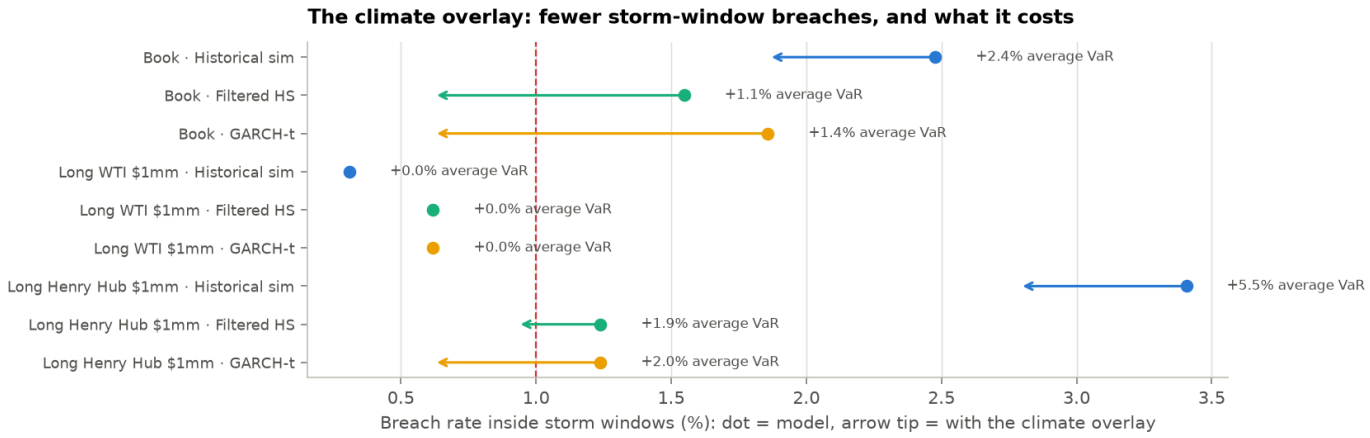
Three 99% one-day VaR models from regime_risk and climrisk.overlay forecast every day from 2010 using data up to the day before: historical simulation (500 days), filtered HS (EWMA 0.94) and GARCH(1,1)-t re-estimated every 250 days. A storm's risk window runs from -2 to +10 trading days around day 0 (hurricane and winter-storm watches give about two days' warning). The overlay multiplies VaR inside a window by k, the ratio of realised to model-implied volatility in earlier storms of the same kind (only days before the window opens), capped at 3 and switched on after 2 past events.

Breach rates (target 1%); 323 storm-window days in the backtest

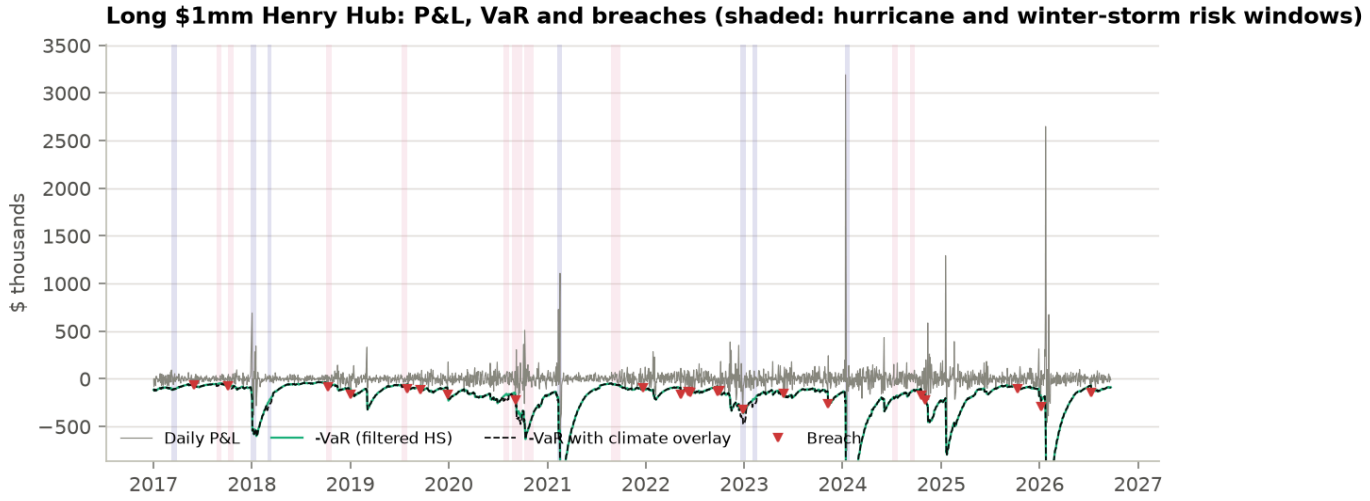
Position	Model	All days	Storm windows	Other days	Fisher p	Storm breaches, then with overlay	Extra avg VaR
Book	Historical sim	1.05%	2.48%	0.94%	0.018	8 -> 6	2.4%
Book	Filtered HS	0.98%	1.55%	0.94%	0.209	5 -> 2	1.1%
Book	GARCH-t	1.17%	1.86%	1.12%	0.174	6 -> 2	1.4%
Long WTI	Historical sim	1.31%	0.31%	1.39%	0.988	1 -> 1	0.0%
Long WTI	Filtered HS	1.05%	0.62%	1.09%	0.866	2 -> 2	0.0%
Long WTI	GARCH-t	1.29%	0.62%	1.34%	0.929	2 -> 2	0.0%
Long Henry Hub	Historical sim	1.08%	3.41%	0.89%	<0.001	11 -> 9	5.5%
Long Henry Hub	Filtered HS	0.87%	1.24%	0.84%	0.306	4 -> 3	1.9%
Long Henry Hub	GARCH-t	0.87%	1.24%	0.84%	0.306	4 -> 2	2.0%



Breach rates inside storm windows (filled) and on other days (hollow).



The overlay: storm-window breach rate before (dot) and after (arrow tip), with its cost in average VaR.



A long Henry Hub position: P&L, filtered-HS VaR, the overlay and breaches.

5. Climate stress tests

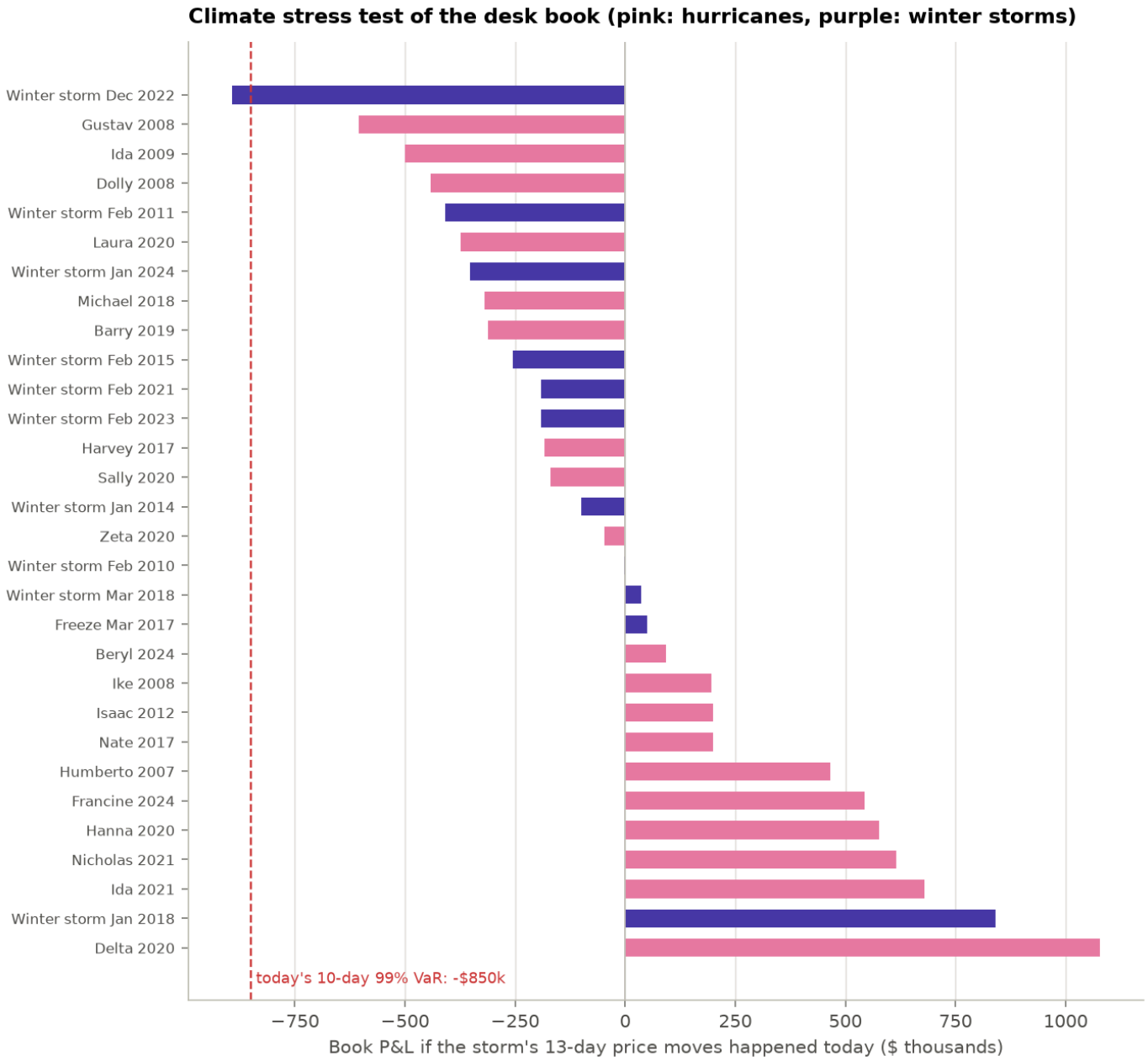
Each event's actual 13-day price changes (day -2 to +10) applied to today's positions: +$4mm WTI, -$2mm Brent, +$2mm HenryHub for the book. The 10-day VaR is the square-root-of-time scaling of today's one-day filtered-HS VaR.

Scenario losses in context

Position	10-day 99% VaR	Worst climate scenario	Event	Scenarios beyond VaR	Worst 13 days, any cause	Ending
Book	$850k	−$893k	Winter storm Dec 2022	1 of 30	−$1.73mm	2020-03-30
Long WTI $1mm	$249k	−$226k	Gustav 2008	0 of 30	−$588k	2020-03-20
Long Henry Hub $1mm	$272k	−$483k	Winter storm Dec 2022	2 of 30	−$894k	2026-02-11

The ten worst scenarios for the book

Event	WTI	Brent	Henry Hub	Book P&L	x 10-day VaR
Winter storm Dec 2022	-0.6%	-4.7%	-48.3%	−$893k	-1.05
Gustav 2008	-22.6%	-24.1%	-9.1%	−$605k	-0.71
Ida 2009	-4.7%	-0.1%	-15.6%	−$500k	-0.59
Dolly 2008	-8.0%	-11.5%	-17.5%	−$441k	-0.52
Winter storm Feb 2011	-1.6%	+6.2%	-11.1%	−$409k	-0.48
Laura 2020	-10.1%	-9.0%	-7.5%	−$374k	-0.44
Winter storm Jan 2024	+8.3%	+6.9%	-27.4%	−$353k	-0.42
Michael 2018	-10.7%	-8.6%	-3.2%	−$320k	-0.38
Barry 2019	-5.7%	-6.2%	-10.4%	−$312k	-0.37
Winter storm Feb 2015	+1.6%	+13.6%	-2.4%	−$256k	-0.30



Book P&L under every historical climate scenario.

Method notes, limitations and references

- Models come from the package `regime_risk` (regime-risk 1.0.0 (installed package)), the tested library of github.com/n9omi/regime-risk-and-model-validation; this project adds only the event logic, the event study and the overlay.
- Spot prices, not futures: spot Henry Hub reacts to same-day weather more than the front-month contract, which exaggerates gas moves relative to what a futures book would see.
- Event dates are physical, not the moment markets learned of them: forecasts move prices days earlier, which the run-up window [-5,-1] and the risk window starting at -2 only partly capture.
- NOAA stopped updating the billion-dollar database in 2025, so winter storms after 2024 are missing, including the late-January 2026 cold snap that produced the worst 13 days for a long gas position.
- 18 and 11 events are small samples: a single storm (February 2021) can move an average, which is why the placebo tests and per-event tables matter more than any one mean.
- The overlay assumes the risk manager knows about two trading days ahead that a named storm will hit; NHC and NWS watches usually give that, but its timing is idealised.

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